

Gym Management System

Case Study

A company has decided to develop a multi-tenant Gym Management System to help gym owners manage their gyms through one platform. The system allows multiple gyms to use the same application while ensuring that each gym can access only its own data.

- * The system supports multiple gyms. Each gym has a unique Gym ID, name, address, phone number, email, status, subscription status, and creation date. Every gym works independently, and the data of one gym cannot be accessed by another gym.

- * A gym owner can register a new gym. The registration request is reviewed by the System Admin. If the request is approved, the system creates a Gym Admin account and allows the gym to continue the subscription process.

- * Before using the system, the Gym Admin must select one Subscription Plan. Each subscription plan has a unique ID, name, duration, price, maximum number of members, maximum number of staff accounts, and status.

- * After selecting a subscription plan, the Gym Admin uploads a payment receipt. The payment is stored as a Gym Payment and reviewed by the System Admin. If the payment is approved, a Gym Subscription is activated and the gym becomes active.

- * The system stores information about users. Each User has a unique ID, full name,

email, password, phone number, account status, and belongs to one gym except the System Admin.

- * Every user is assigned one Role. The system supports the following roles: System Admin, Gym Admin, Trainer, Receptionist, and Member.

- * Each gym can have many trainers and receptionists. Every trainer and receptionist belongs to only one gym.

- * Each gym can create one or more Memberships. Every membership has a unique ID, name, duration, price, and status and belongs to only one gym.

- * Members register in one gym by selecting a membership and uploading a payment receipt. Member payments are stored in the Member Payment entity until they are reviewed by the Gym Admin.

- * The Gym Admin reviews member information and payment before approving or rejecting the membership request. Approved members become active and can use the gym services..

- * Each member belongs to only one gym and has one active membership at a time.

- * Each trainer can train many members, while each member is assigned to one trainer at a time.

- * The system records attendance for members. Each Attendance record stores the member, attendance date, check-in time, check-out time, attendance status, and the receptionist who recorded the attendance.

* Members can submit Freeze Requests to temporarily freeze their memberships. Each freeze request belongs to one member and one membership. The Gym Admin reviews the request before approving or rejecting it.

* The system stores financial operations using the Transaction entity. Each transaction belongs to one gym and contains the transaction type, amount, payment method, category, and transaction date.

* The system stores notifications for users. Each Notification belongs to one user and is used to inform users about important events such as gym approval, payment approval, membership approval, or freeze request decisions.

* The system maintains an Activity Log. Every important operation performed by a user is recorded with the action type, affected record, timestamp, and the user who performed the action.

* The System Admin can manage all gyms in the platform, while each Gym Admin can manage only the users, memberships, attendance records, payments, and transactions that belong to their own gym.